DESIGN AND ANALYSIS OF ALGORITHMS

COURSE INTRODUCTION

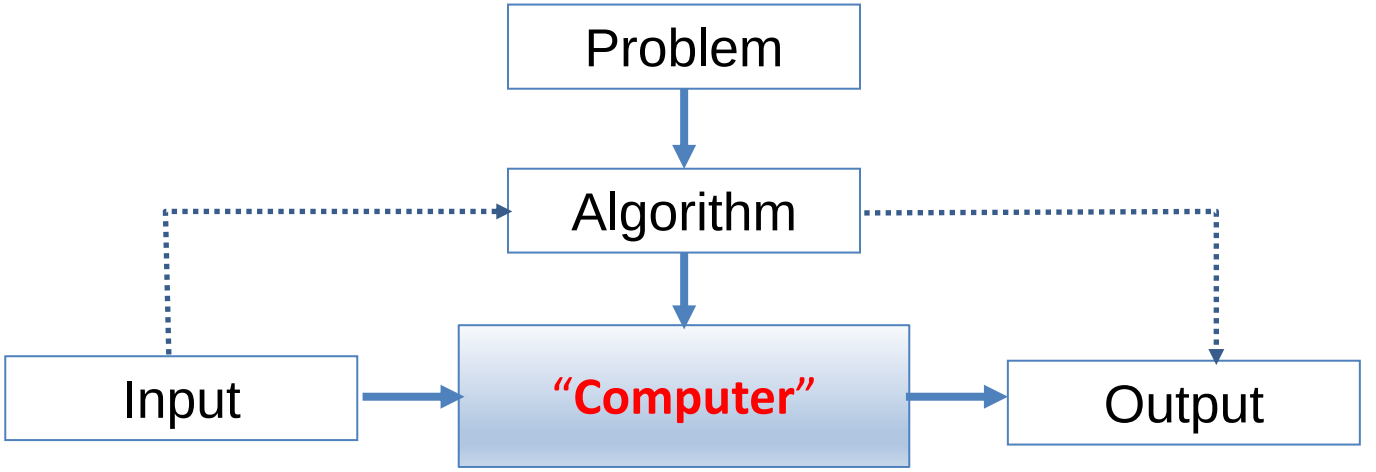
minhnth@hus.edu.vn

Course overview

From: internet

Course overview

- Why we need the algorithms?
  - Problems in life.
  - Problems in science.
  - In Computer and Information Science.



Course overview

- Emphasizes on **Design** and **Analysis** of Algorithms:
  - Design Algorithm:
    - How to make algorithm for solving a given problem?
  - Analysis of Algorithm
    - The designed algorithm is good or not?
    - Benchmark: time and resource.

Course overview

- **Topics covered**
  - ***Theoretical foundations:*** Turing Machines, Computability, and Complexity Classes (P, NP, NP-Complete).
  - ***Design Strategies***: Recursive; Divide and Conquer; Trial and Error; Greedy; Dynamic Programming.
  - ***Algorithm Analysis***: Mathematical proofs (asymptotic notations), empirical evaluation (running experiments).
  - ***Problem Solving***: Transforming real-world problems into algorithmic models.

Objective

Main goals of this course:

- ***Knowledge***: Know how to make problem specification; grasp the basic techniques for algorithm design and assess their complexities.
- ***Skills***: Ability to build algorithm for specific problems, give the complexity of the algorithm.
- ***Attitude***: Having a serious learning attitude, ability to self-study and thinking in developing application for real problems.

DSA vs. DAA: the difference

Feature	Data Structures & Algorithms	Design & Analysis of Algorithms
Focus	Data Organization, Standard Algorithms	Problem-Solving Strategies, Performance Analysis
Key Question	"How to store data efficiently?" (Arrays, Trees, Graphs, Hash Tables)	"How to solve a problem efficiently?" (Divide & Conquer, Greedy, DP)
Tools	Implementation details, Pointers, Memory management…	Mathematical proofs, Asymptotic notation (O,Ω,Θ), Recurrences…
Outcome	A working program using correct structures	An efficient, optimized, and provable algorithm
Level	Understand & Apply (Knowledge)	Analyze, Evaluate & Create (Wisdom/Design)

"DSA provides the tools-The Tools; DAA engineers the solution-The Strategy."

Lecture schedules

- Semester 2, AY 2025-2026
- Credits: 3 (150h= 30 class + 30 lab + 90 self-study)
- Class: Tuesday, 8:50-10:40, room 102-T4

  Thursday, 8:50-10:40, room 103-T4

  Thursday, 10:45-12:30, room 103-T4

  Lab: Computer room - T5
- Course language: Vietnamese, English

Assessments

- Weekly (homework, attendance): 20%
  - Submit and Cross Assessment weekly homework
  - Check attendance every class
- Midterm (homework, test): 20%
  - Weekly homework mark
  - Midterm test
- Final (exam, project): 60%
  - Group project / Final test

Lecturer and tutor

- Lecturer
  - Assoc.Prof. Nguyễn Thị Hồng Minh
  - 0904101065 (phone, zalo, whatsapp)
  - minhnth@hus.edu.vn, minhnth@gmail.com
- Tutors
  - MSc. Trần Bá Tuấn
    (032.745.7300, tranbatuan@hus.edu.vn)
  - BSc. Trần Anh Minh
    (0949559640, trananhminh@hus.edu.vn)

Course resources

- Course materials

  Google Classroom

  Google Drive

- Discusions

  Group zalo

  Email: minhnth@hus.edu.vn

  Subject begin with: [MAT3552/MAT3504].<subject>

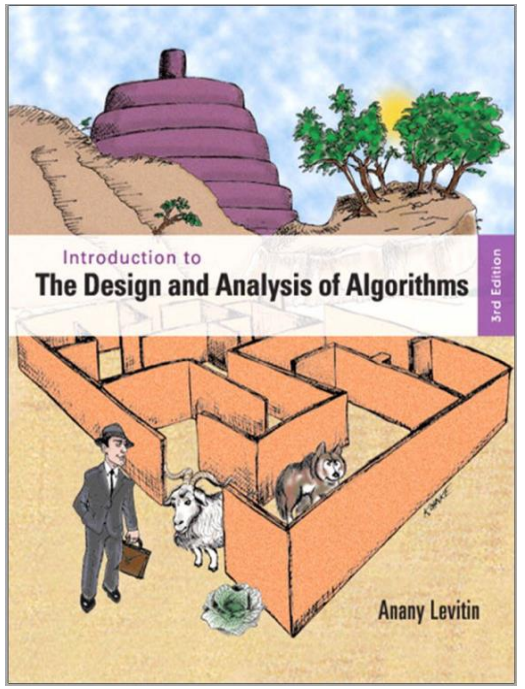
- Software

  Programming language: **Java, C++, Python**

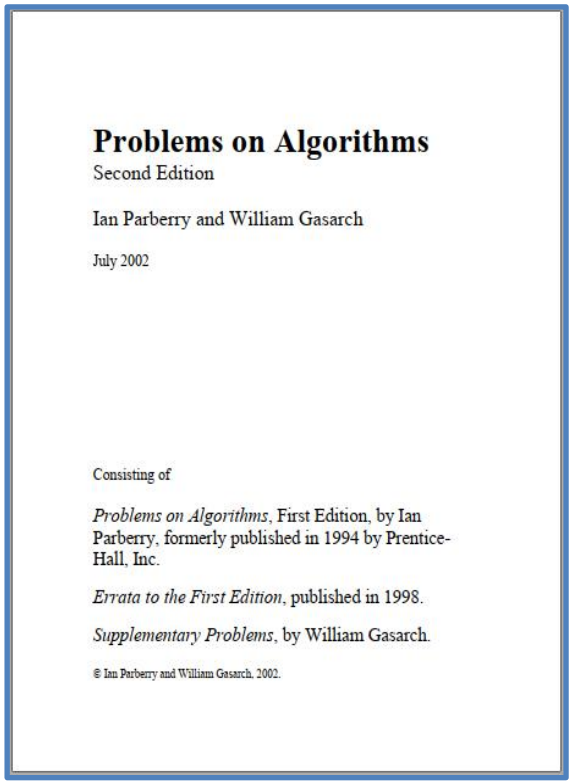
  Interactive app: Google Form, Kahoot! (smart phone, laptop)

Course resources

- Reference books:



- The Design and Analysis of Algorithms (3rd), Anany Levitin, Pearson, 2012



Problems on Algorithms
Second Edition

Ian Parberry and William Gasarch

July 2002

Consisting of

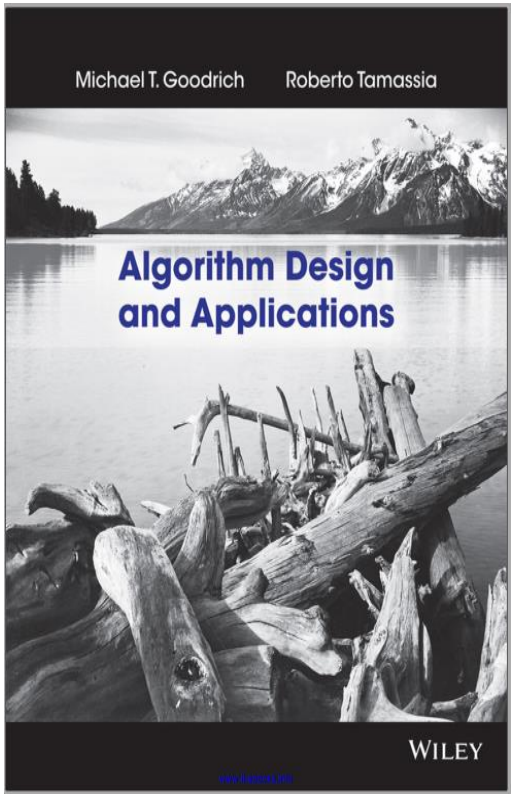
Problems on Algorithms, First Edition, by Ian Parberry, formerly published in 1994 by Prentice-Hall, Inc.

Errata to the First Edition, published in 1998.

Supplementary Problems, by William Gasarch.

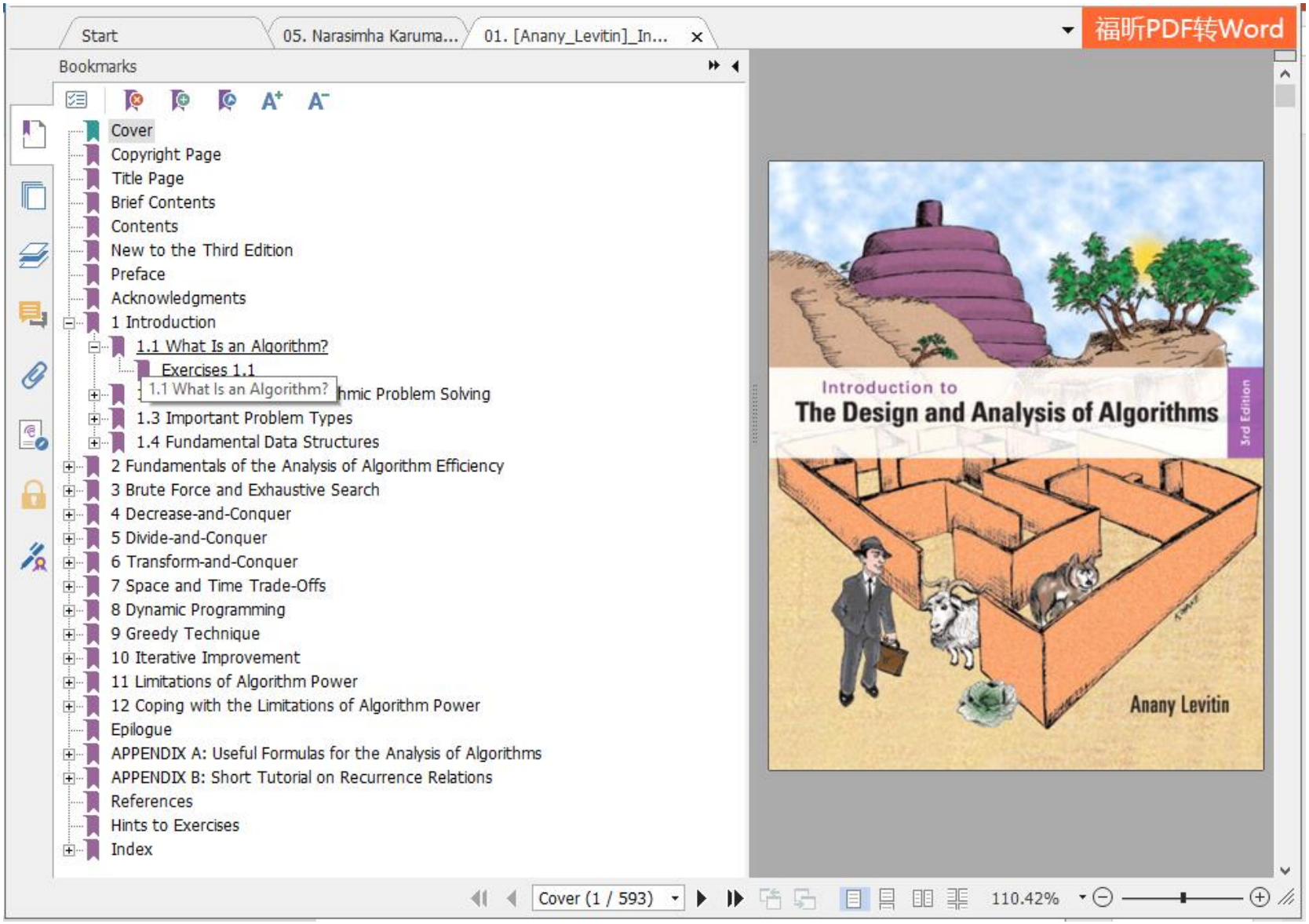
© Ian Parberry and William Gasarch, 2002.

- Problems on Agorithms, Ian Parberry and W. Gasarch, ebook, 2002



- Algorithm Design and Applications, Michael T. Goodrich, R. Tamassia, 2015

Course resources



Class Rules (Nguyên tắc lớp học)

- Đi học đúng giờ
- Không nói chuyện riêng, nói chuyện điện thoại trong lớp.
- Vệ sinh bảng, phấn. Chuẩn bị máy chiếu.

- Điểm danh hàng ngày, tham dự 80% giờ học để dự đánh giá cuối kì.
- NGHIÊM CẤM GIAN LẬN trong làm bài tập, kiểm tra. Đình chỉ môn học khi phát hiện gian lận.

Expected results

- Complete the course with the best score;
- Subject material folder: lectures, documents, assignments;
- Ideas for projects, applications;
- Memorable classroom experience.
- …

From: https://www.insightoftheday.com/motivational-quote-by-robert-collier-10-31-2019

Thank you for going together!

Assoc. Prof. Nguyen Thi Hong Minh

Department of Informatics

Faculty of Mathematics, Mechanics and Informatics

VNU University of Science

Bộ môn Tin học

Khoa Toán - Cơ - Tin học

Trường Đại học Khoa học Tự nhiên

Đại học Quốc gia Hà Nội

minhnth@hus.edu.vn